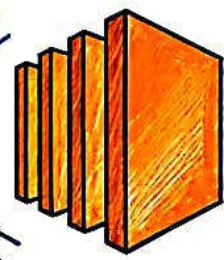


MASTER THE HEART OF AWS

EC2



Amazon
Elastic
Compute
Cloud



Ever wondered where your website, app,
or backend actually runs in the cloud?

The Answer is



EC2



A virtual server
in the cloud
that gives you
✓ Compute Power
✓ Flexibility
✓ Control
✓ Scalability

In this series, you will learn

-  What is EC2?
-  Architecture
-  How to Launch
-  Components
-  Security & Networking
-  Storage (EBS)
-  User Data & Metadata
-  Scaling, ASG & Load Balancer

-  Instance Types
-  Pricing & Right Choice
-  Real World Architecture
-  Troubleshooting
-  Interview Concepts
- & much more...



From Basics to Real World
Let's become EC2 Expert!



Amazon EC2

2

What is EC2 ?



Amazon **EC2** (Elastic Compute Cloud) is a web service that provides secure, resizable compute capacity in the cloud. In simple words, **EC2** gives you **virtual servers (instances)** that you can use anytime, from anywhere.

Think of EC2 Like This :



You need a server to run your website / app / backend.

EC2 gives you that server in the cloud.

Key Point :

- ☆ You get a virtual computer in the cloud.
- ☆ You can install any OS, software and run your applications.
- ☆ You **pay only for what** you use.
- ☆ You can start, stop, resize or terminate anytime.

Why Do We Need EC2 ?

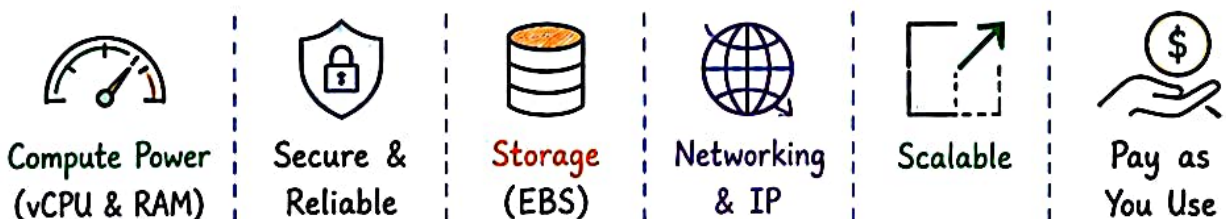
- ✓ To host websites, applications, and APIs.
- ✓ To run **backend services** and databases.
- ✓ To test and develop in the cloud.
- ✓ To scale easily when traffic increases.
- ✓ To reduce cost and maintain **high availability**.



In One Line :

EC2 gives you a virtual server in the cloud that you can control, **scale**, and use as per your need.

What You Get With EC2 ?



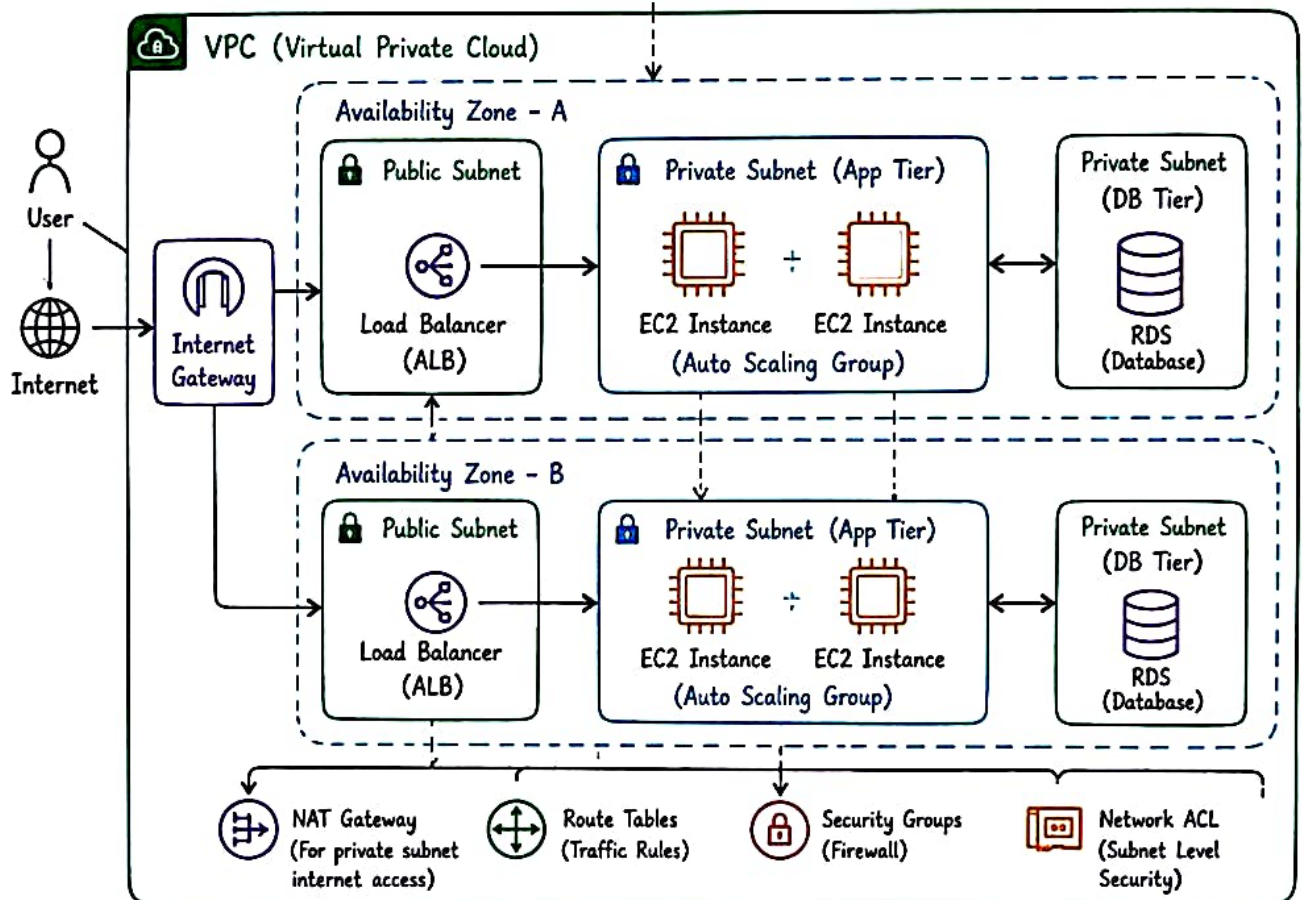
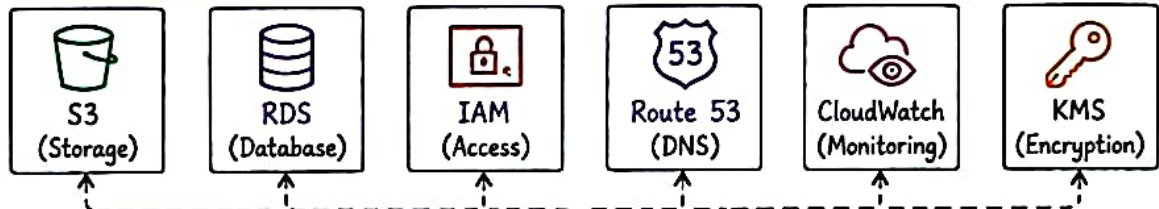
EC2 is the building block of most applications in **AWS**.

EC2 ARCHITECTURE

3

EC2 works with many AWS services together.
Together they give you a secure, scalable and reliable environment.

AWS SERVICES THAT WORK WITH EC2



How it works (Flow)

- ① User sends request through Internet.
- ② Request goes to Internet Gateway.
- ③ Traffic reaches Application Load Balancer (ALB) in Public Subnet.
- ④ ALB sends request to **EC2 Instances** in Private App Subnet.
- ⑤ EC2 talks to RDS in Private DB Subnet for data.
- ⑥ Response goes back to user.

Key Points

- ★ EC2 instances are placed in Private Subnets for security.
- ★ ALB is in Public Subnets to receive traffic from Internet.
- ★ RDS is in Private DB Subnets, not exposed to Internet.
- ★ **Security Groups** act as Firewall at instance level.
- ★ **Route Tables** control where traffic goes.
- ★ Multi-AZ gives High Availability.

★ EC2 is not just a server, it is the **heart** of your cloud architecture.

HOW TO LAUNCH EC2 INSTANCE

4



Launching an EC2 instance means creating your own **virtual server** in the cloud in just a few minutes.



STEP-BY-STEP PROCESS

1



Choose an AMI

- AMI (Amazon Machine Image) is the OS and software template.
- You can use Amazon Linux, Ubuntu, Windows or your own AMI.

2



Choose an Instance Type

- Select the right size (vCPU, RAM) as per your workload.
- **Example:** t3.micro for small apps, t3.large for heavy workloads.

3



Configure Network

- Choose VPC and Subnet.
- Assign Public IP if you need Internet access.
- Select or create **Security Group**.

4



Configure Storage (EBS)

- Set the root volume size (e.g. 20 GB).
- Choose volume type (gp3, io2, st1 etc.).
- Add more volumes if required.

5



Add Key Pair

- Key pair is used to SSH into your instance.
- Choose existing key pair or create a new one (**.pem** file).

6



Configure Advanced Settings (Optional)

- IAM role, User data (script), Monitoring, Placement group, etc.

7



Review & Launch

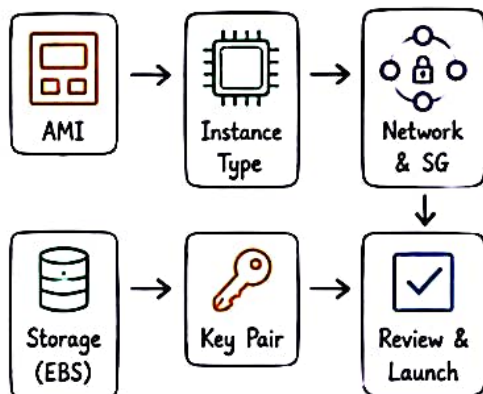
- Review all settings.
- Click "**Launch Instance**".

In Short

1. Pick an AMI
2. Choose Instance Type
3. Configure Network
4. Configure Storage
5. Add Key Pair
6. Advanced Settings (opt.)
7. Review & Launch

Your EC2 is Ready!

LAUNCH FLOW



★ Before You Launch, Make Sure:

- ✓ You are in the right Region.
- ✓ You have the correct AMI.
- ✓ Security Group allows the required ports.
- ✓ Key pair is downloaded and safe.
- ✓ Storage size is enough for your app.



A few clicks today,
Your **virtual server** is ready to power your ideas!

EC2 COMPONENTS

5

EC2 is made of many building blocks.

Understanding them helps you launch, manage and secure your instances the right way.

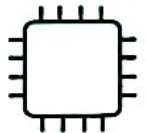


1 AMI (Amazon Machine Image)



- Blueprint of your instance.
- Contains OS, applications and configurations.
- You can use AWS AMI, Marketplace AMI or your own Custom AMI.

2 Instance Type



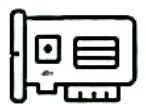
- Defines vCPU, Memory, Storage and Network performance.
- Example: t3.micro, m5.large etc.

3 VPC & Subnet



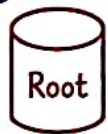
- VPC is your private network in AWS.
- Subnet is a range of IP addresses within VPC.

4 Network Interface (ENI)



- Provides network connectivity to the instance.
- Primary ENI is created by default.

5 Root Volume



- The primary EBS volume.
- Contains OS and boot files.
- Deleted when instance is terminated (by default).

6 User Data



- Script or commands that run when instance starts for the first time.
- Useful for automation like installing packages, starting services etc.

7 Security Group



- Acts as a virtual firewall for your instance.
- Controls inbound and outbound traffic.
- It is stateful.

8 Key Pair



- Used to securely connect to your instance.
- Consists of Public Key (AWS) and Private Key (.pem) (you).

9 Elastic IP (EIP)



- Static public IP address.
- Remains the same even if you stop/start instance.
- Useful for SSH, Websites etc.

10 EBS Volume



- Persistent block storage.
- Attached to EC2 instance.
- Data remains even if instance is stopped.

11 Instance Metadata



- Data about your instance.
- Accessible from inside the instance.
- Example: Instance ID, AZ, MAC address, User data etc.

12 Placement Group (Optional)

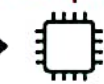


- Logical grouping of instances.
- Types: Cluster, Spread, Partition.
- Helps in low latency and high performance apps.

How These Components Work Together ?



AMI



Instance Type



VPC / Subnet



Security Group



EBS



Key Pair



EC2 Instance



Each component plays an important role in making your EC2 instance secure, available, scalable and cost-effective.





KEY PAIR & SSH to EC2

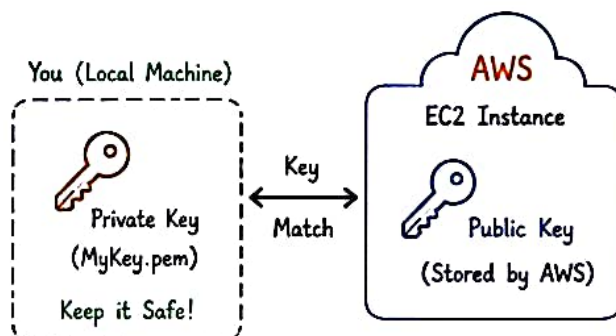
6

Key Pair is used to securely connect to your EC2 instance.
 It uses public-private key cryptography.
 You keep the private key, AWS stores the public key on the instance.

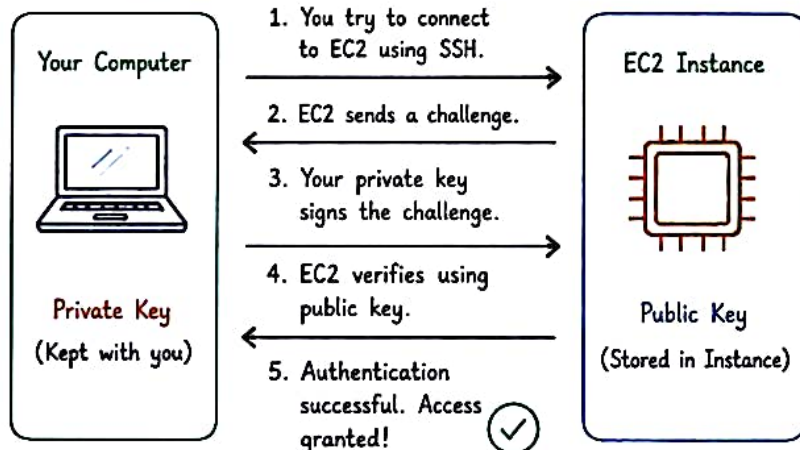


1 What is a Key Pair ?

- A key pair has two parts - Public Key and Private Key.
- AWS stores the Public Key on the EC2 instance.
- You download and keep the Private Key (.pem file) on your local machine.



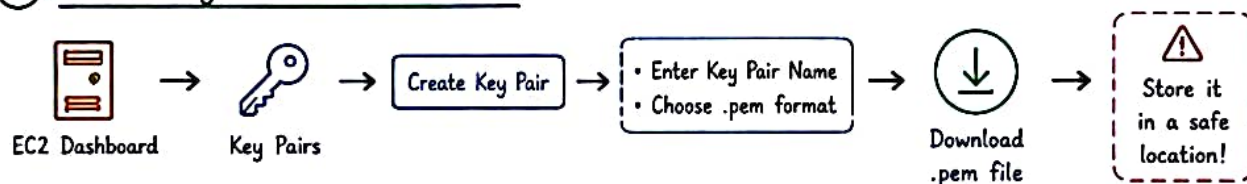
2 How SSH Works ?



3 Important Points

- ★ Never share your private key (.pem) with anyone.
- ★ If someone gets your private key, they can access your instance.
- ★ Use proper file permission for .pem file. (chmod 400 MyKey.pem)
- ★ One key pair can be used with multiple instances.

4 Create Key Pair in AWS Console



5 SSH to EC2 Instance (Linux Example)

1. Set permission for .pem file
`chmod 400 MyKey.pem`
 ↳ Only you can read this file.

2. Connect using SSH

```
ssh -i MyKey.pem ec2-user@<Public-IP>
```

- ec2-user : default user for Amazon Linux
- Replace <Public-IP> with your instance IP

Example :

```
$ chmod 400 MyKey.pem
$ ssh -i MyKey.pem ec2-user@3.108.25.67

[ec2-user@ip-10-0-1-25 ~]$
```

Amazon Linux 2023
<https://aws.amazon.com/linux/amazon-linux-2023>

You are connected!



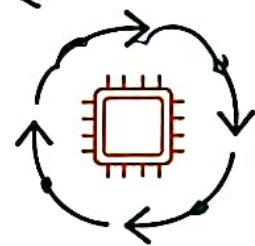
Key Pair + SSH = Secure and Encrypted way to access your EC2 instance.
 Always keep your private key safe!



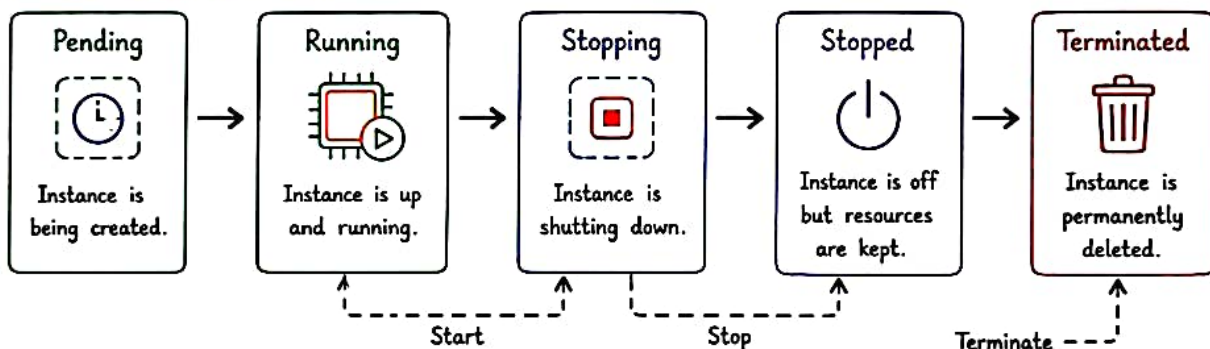
EC2 LIFECYCLE

7

An EC2 instance goes through different states in its life. Understanding lifecycle helps you manage cost, availability and resources smartly.



1. EC2 Lifecycle States



2. What Happens to Different Things?

Component	Running	Stopped	Terminated
EC2 Instance (Compute)	Instance is ON. You are billed .	Instance is OFF. You are NOT billed for compute.	Instance is deleted permanently. You are NOT billed.
EBS (Attached Volumes)	Data is safe. You are billed for EBS storage.	Data is safe. You are billed for EBS storage.	Deleted by default. (If "Delete on Termination" is enabled)
Public IP (If Auto-assign Public IP)	Public IP is assigned.	Public IP is released. (changes when you start again)	Public IP is released.
Elastic IP (EIP)	Remains with you. (Until you release)	Remains with you. (Until you release)	Remains with you. (Until you release)
Key Pair	Remains with you.	Remains with you.	Remains with you.
Security Group	Remains (your configuration saved)	Remains.	Remains.
ENI & Private IP	ENI attached. Private IP assigned.	ENI detached. Private IP released.	ENI deleted. Private IP released.

3. Key Points

- ☆ Use Stop/Start to save cost on compute. (Good for dev/test environments)
- ☆ Use **Terminate** when you no longer need the instance.
- ☆ EBS charges apply even when instance is stopped.
- ☆ Data on EBS is safe during Stop/Start.
- ☆ Always take a Snapshot before terminating important instances.

4. Quick Summary

	Running	=	On	=	You are billed
	Stopped	=	Off	=	You save on compute
	Terminated	=	Deleted	=	Resources gone (by default)



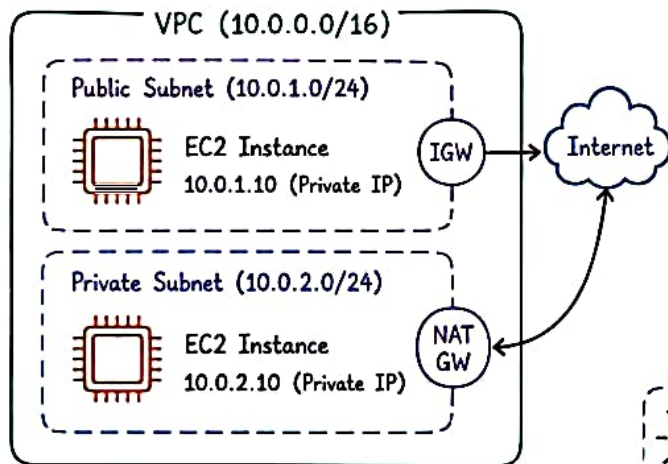
Choose the right action at the right time to **save cost** and protect your data.

EC2 NETWORKING

8

Every EC2 instance needs network to communicate.
Networking in AWS is built inside a VPC (Virtual Private Cloud).

1. Basic Networking for EC2

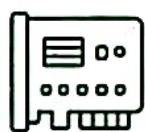


2. IP Addresses in EC2

- Private IPv4 (default)
Assigned from your Subnet CIDR.
Example: 10.0.1.10
- Public IPv4 (Optional)
Used to access instance from Internet.
Can be Auto-assigned or Elastic IP.
- IPv6 (Optional)
Supported in dual-stack VPC.
Provides IPv6 address.

For security, keep instances in private subnet and access through NAT, Bastion Host or Load Balancer.

3. Network Interfaces (ENI)



- Each EC2 instance has at least one Elastic Network Interface (ENI).
- ENI has Private IP, MAC address and security group.
- You can attach multiple ENIs to an instance.

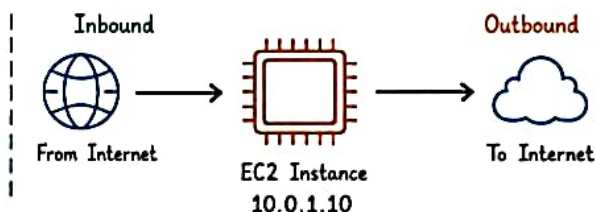
4. Elastic IP (EIP)



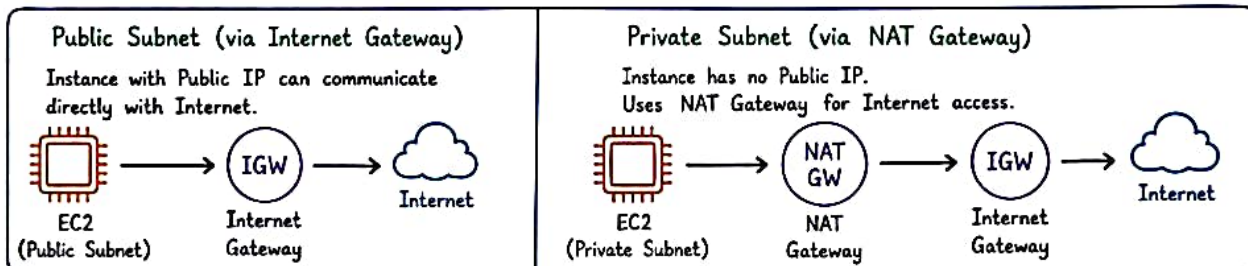
- Static public IP address.
- Remains the same even if you stop/start instance.
- Useful for SSH, Websites, APIs etc.
- Associate it with your instance.

5. Types of Traffic

- Inbound Traffic: Comes from outside to your instance.
- Outbound Traffic: Goes from your instance to outside.
- By default, all outbound traffic is allowed.
- Control using Security Groups and NACL.



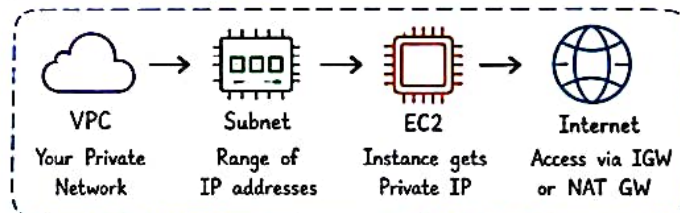
6. How Internet Access Works



7. Important Points

- ★ Private IP changes when instance is stopped.
- ★ Public IP changes on stop/start (if auto-assigned).
- ★ Use Elastic IP for static access.
- ★ Place instances in correct subnet for security.
- ★ Use Security Groups to control traffic.

8. Quick Summary



Good networking design makes your EC2 instance Secure, Available and Scalable.





EC2 SECURITY

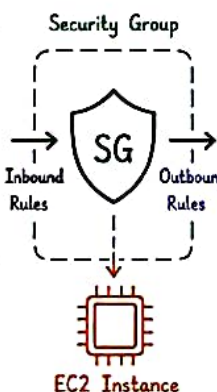
9

Keep your instances **SAFE**, **ACCESSIBLE** and **CONTROLLED**.
In EC2, security mainly depends on Security Groups and Key Pairs.

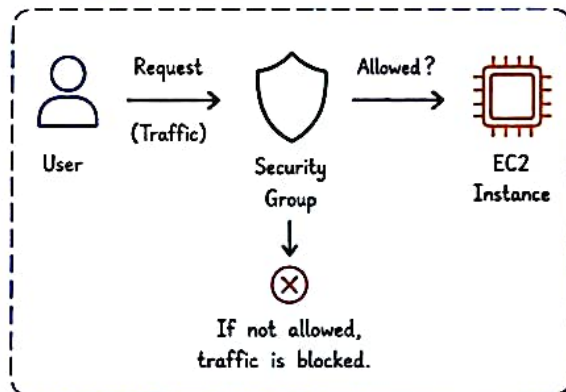


1. What is a Security Group ?

- Security Group acts as a **virtual firewall** for your EC2 instance.
- It controls inbound and outbound traffic.
- It is **stateful** → return traffic is allowed automatically.
- Rules are **Allow** by default (**Deny** all if no rule matches).



2. Security Flow



3. Important Ports & Protocols

Port	Protocol	Service / Use	Description
22	TCP	SSH	Secure Shell access (Linux).
3389	TCP	RDP	Remote Desktop access (Windows).
80	TCP	HTTP	Web traffic (Not secure).
443	TCP	HTTPS	Secure Web traffic (SSL/TLS).
21	TCP	FTP	File Transfer Protocol.
25	TCP	SMTP	Email sending.
53	TCP/UDP	DNS	Domain Name System.
3306	TCP	MySQL	MySQL Database.
5432	TCP	PostgreSQL	PostgreSQL Database.
6379	TCP	Redis	Redis Cache.
8080	TCP	HTTP-Alt	Common for Web Apps / Proxy.

4. Inbound vs Outbound Rules

Inbound Rules



- Control traffic coming into your instance.
- Example: Allow SSH from your IP.

Outbound Rules



- Control traffic going out from your instance.
- By default, all outbound traffic is allowed.

5. Example - Allow SSH (Port 22) from My IP

Type	Protocol	Port Range	Source	Description
SSH	TCP	22	203.0.113.10/32	My IP for SSH Access

★ Use /32 for a single IP. Use 0.0.0.0/0 to allow from anywhere (Not recommended).

Best Practices

- ✓ Allow only required ports.
- ✓ Use specific IP ranges.
- ✓ Use least privilege access.
- ✓ Review rules regularly.



6. Key Pair vs Security Group

Key Pair



- Used to authenticate (login).
- Works at OS level.
- Example: SSH to EC2.

Security Group



- Used to authorize (network access).
- Works at network level.
- Example: Allow port 22.

7. Quick Summary

- ★ Security Group = Firewall for EC2.
- ★ It is stateful.
- ★ Control inbound & outbound traffic.
- ★ Allow only what is needed.
- ★ Keep your instances secure!



Good security today, safe infrastructure tomorrow.





EC2 STORAGE

10



Storage is a critical part of EC2.

EC2 uses EBS for persistent storage and **Instance Store** for temporary, high-performance storage.

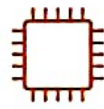
1. Storage Options in EC2

EBS (Elastic Block Store)



- Persistent block storage.
- Data remains even if instance stops.
- Can be attached to multiple instances (one at a time).
- Backed by AWS infrastructure.

Instance Store

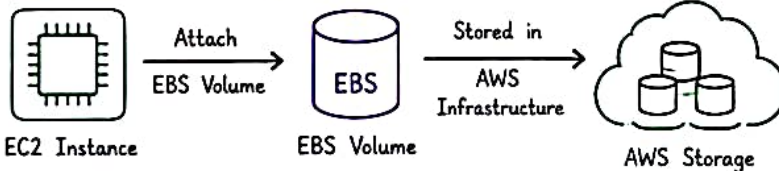


- Temporary storage (on physical disk).
- Data is lost when instance stops, terminates or fails.
- Very high I/O performance.

2. EBS Volume Types

Volume Type	Best For	Key Features
gp3	General purpose workloads	• Balanced price & performance • Baseline 3,000 IOPS & 125 MB/s • Can scale up to 16,000 IOPS and 1,000 MB/s
gp2	General purpose (older)	• Baseline performance • IOPS depends on volume size (min 100, max 16,000)
io1 / io2	I/O intensive critical workloads	• Highest performance • Consistent low latency • io2 provides higher durability
st1	Throughput intensive workloads	• Low cost HDD • High throughput, not for IOPS • Examples: Data lake, Logging
sc1	Cold workloads (less frequent access)	• Lowest cost HDD • For infrequent access data • Examples: Backups, Archives

3. How EBS Works



💡 You attach an EBS volume to an EC2 instance. The data is safely stored in AWS infrastructure.

4. EBS vs Instance Store

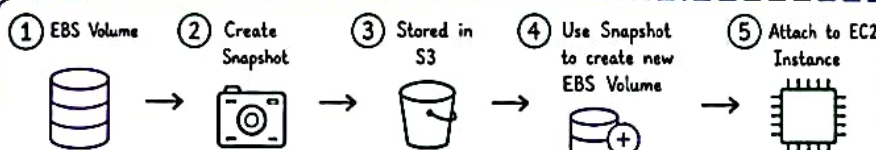
Feature	EBS	Instance Store
Persistence	Yes	No
Lives on	AWS Infrastructure	Physical Server
Performance	Good (varies by type)	Very High
Durability	High (replicated)	Low (ephemeral)
Use case	Databases, Apps, Critical Data	Cache, Temp Data, Buffer, Logs
Cost	Charged (per GB)	Included (no extra cost)
Attach/Detach	Yes	No

5. Key Points

- ★ Use EBS for important data that must be safe.
- ★ Use Instance Store for temporary and high-speed data.
- ★ Choose the right EBS type based on your workload.
- ★ You can take EBS Snapshot for backup.
- ★ EBS is AZ specific (can't attach across AZs).
- ★ Always monitor EBS performance (IOPS & Throughput).



6. EBS Snapshot (Backup)



💡 Snapshots are incremental, stored in S3 and can be used for backup or to create new volumes.



Right storage choice = Better Performance, Lower **Cost** and Data Safety.

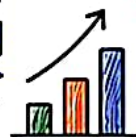




EC2 MONITORING & LOGGING

11

Monitoring helps you know the health, performance and activity of your EC2 instances in real-time.



1. Why Monitor EC2 ?

- ✓ Know if your instance is healthy.
- ✓ Track CPU, Memory, Disk, Network usage.
- ✓ Find issues before users face them.
- ✓ Optimize performance and cost.
- ✓ Maintain security and compliance.

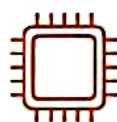


AWS CloudWatch is the main service used to monitor EC2.

2. What Can You Monitor ?

Category	What It Monitors	Examples
Compute	CPU utilization, Credit usage	CPUUtilization, CPUCreditBalance
Memory	Available memory, Swap	MemoryAvailable, SwapUsage
Storage	Disk usage, Disk I/O	DiskReadOps, DiskWriteBytes
Network	Network in/out, Packets	NetworkIn, NetworkOut
System	Status check, Events	StatusCheckFailed, SystemStatus

3. Monitoring Architecture with CloudWatch

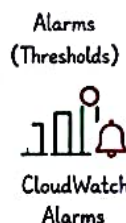


EC2 Instance

Metrics
(CPU, Disk,
Network etc.)



Amazon
CloudWatch



Actions
(Notify / Auto
Action)



Email



Lambda



SNS



Automation

4. Key CloudWatch Metrics (Common)

Metric Name	Description	Good Practice
CPUUtilization	CPU usage in %	Keep < 70% for good performance
NetworkIn	Network in (bytes)	Monitor for spikes
NetworkOut	Network out (bytes)	Monitor for spikes
DiskReadBytes	Disk read (bytes)	High values = heavy read
DiskWriteBytes	Disk write (bytes)	High values = heavy write
StatusCheckFailed	Instance health check	Should be 0
CPUCreditBalance	t2/t3 credit balance	Should not be 0

5. EC2 Status Checks

- System Status Check - Checks underlying AWS infrastructure (e.g., hardware, network).
- Instance Status Check - Checks OS, network connectivity, and instance-level issues.

Status	Meaning
○ OK	Everything is healthy.
⚠ Impaired	Some issues detected.
⊗ Insufficient Data	Not enough data yet.
⊗ Failed	There is a problem.

6. Logs from EC2

- You can collect logs from your EC2 using:

- ☆ Amazon CloudWatch Logs Agent
- ☆ syslog / rsyslog
- ☆ Custom scripts



Common Logs

/var/log/messages
/var/log/secure
/var/log/cloud-init.log
/var/log/dmseg

- Send logs to CloudWatch Logs for:

- ✓ Centralized logging
- ✓ Search and analysis
- ✓ Retention & compliance
- ✓ Real-time troubleshooting



EC2
Instance



CloudWatch
Logs

7. Key Takeaways



Monitor
continuously

Always keep an eye on
your instance health
and performance.



Set Alarms

Create CloudWatch
Alarms with
thresholds and
get notified.



Use Logs

Centralize logs in
CloudWatch for
better visibility and
troubleshooting.



Act on Insights

Analyze metrics and
logs to improve
performance and
reduce cost.



Stay Secure

Monitor for unusual
activity and maintain
security at all times.



Monitoring is not just about seeing problems - it's about preventing them!





EC2 SCALING with ASG & LB

12

Scaling and Load Balancing make your application Highly Available, Fault Tolerant and able to handle **more traffic**.



1. Why Scaling & Load Balancing?

- ☆ Traffic can go up and down.
- ☆ One server can become overloaded.
- ☆ Instances can fail anytime.
- ☆ We need to handle **more users** without downtime.



ASG adds/removes instances automatically.
LB distributes traffic evenly across healthy instances.

2. What is Auto Scaling Group (ASG)?

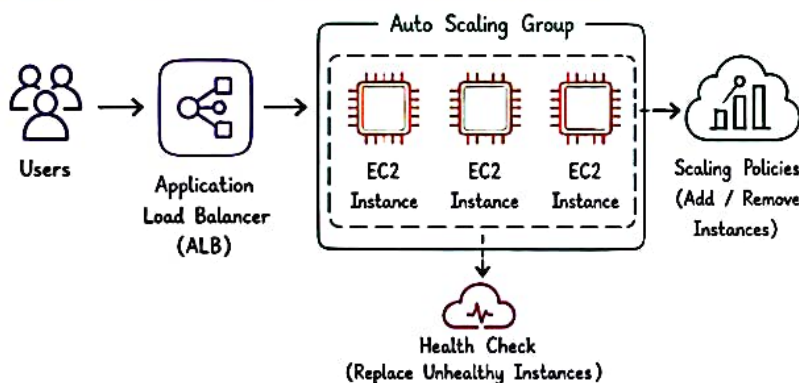
- ASG automatically adjusts the number of EC2 instances based on demand.
- Ensures the right number of instances are running at all times.



ASG Components:

- ✓ Launch Template / Configuration
- ✓ Desired Capacity (min, max, desired)
- ✓ Scaling Policies (Target tracking / Step scaling)
- ✓ Health Checks & Replacement

3. How ASG & Load Balancer Work Together



4. Scaling Policies

Policy Type	What It Does
Target Tracking	Keeps a metric (like CPU) at a target value. Scales in/out automatically.
Step Scaling	Scales based on metric breaching specific thresholds.
Simple / Scheduled Scaling	Scale manually or on a fixed schedule.

5. What is Load Balancer (ALB)?

- ALB distributes incoming traffic across multiple targets (EC2 instances).
- It improves Availability, Performance and Fault Tolerance.
- Performs Health Checks on targets and sends traffic only to healthy ones.

ALB Benefits

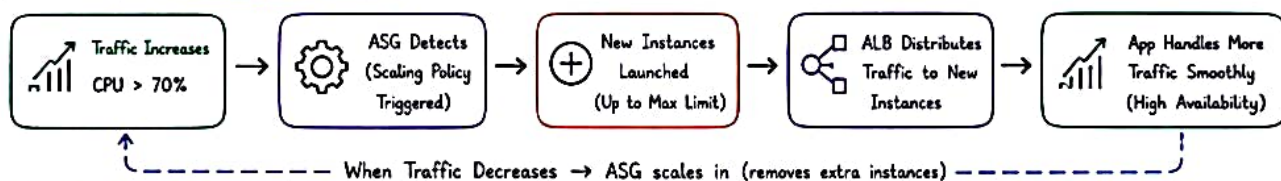
- ✓ High Availability
- ✓ Even Traffic Distribution
- ✓ SSL/TLS Termination
- ✓ Path-based Routing
- ✓ Health Checks

6. Benefits of ASG + LB Together

- ☆ High Availability → No single point of failure.
- ☆ Fault Tolerance → Unhealthy instances are replaced.
- ☆ Scalability → Handles traffic spikes easily.
- ☆ Cost Optimization → Run only what you need.
- ☆ Better Performance → Traffic distributed evenly.



7. Example - Scaling in Action



8. Best Practices

- ✓ Use ALB with ASG for Web Applications.
- ✓ Set proper Min, Max and Desired Capacity.
- ✓ Use Target Tracking for simple scaling.
- ✓ Keep Launch Template updated with latest AMI.
- ✓ Enable Health Checks at ASG & ALB level.
- ✓ Always test scaling with load.
- ✓ Monitor metrics and set correct thresholds.



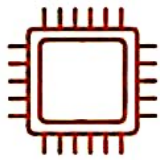
9. Quick Summary

Component	Purpose	Key Benefit
ASG	Automatically adjusts number of EC2 instances	Scalability, Cost Savings, High Availability
ALB	Distributes traffic across multiple instances	Performance, Fault Tolerance, Health Checks
Together	Work as a strong combo	Highly Available, Scalable, Reliable Application



Scale Automatically. Balance Traffic Smartly. Keep Your App **Always Available!**





EC2 INSTANCE TYPES

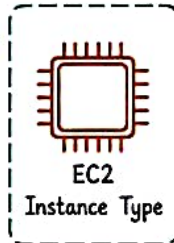
13

Choose the right EC2 instance type to get the best Performance, Cost Efficiency and Scalability.



1. What is an Instance Type ?

- EC2 instance type defines the CPU, Memory, Storage and Network capacity of your instance.
- You can choose the type based on your application requirement.



2. Instance Type Naming (Example: t3.micro)

t3.micro

Family

- t = General Purpose
- m = Memory Optimized
- c = Compute Optimized
- r = Memory Intensive

Size

- nano (Smallest)
- micro
- small
- medium
- large ... (Bigger)

3. Main Instance Families

Family	Best For	Use Case
General Purpose (t, m)	Balanced CPU + Memory	Web servers, App servers, Small to medium apps
Compute Optimized (c)	High CPU performance	Batch processing, Gaming, Video encoding
Memory Optimized (r, x)	Large memory workloads	Databases, Caches, Big data, Analytics
Storage Optimized (i, d, h)	High storage and I/O	NoSQL DB, Data lakes, Hadoop, Log processing

4. Popular Instance Types

Instance Type	vCPU	RAM	Use Case
t3.micro	2	1 GB	Learning / Testing (Free Tier)
t3.small	2	2 GB	Small applications
t3.medium	2	4 GB	Web / App servers
m5.large	2	8 GB	General purpose production apps
c5.large	2	4 GB	Compute intensive workloads
r5.large	2	16 GB	Memory intensive workloads

5. How to Choose the Right Instance Type ?

- ✓ Understand your application requirement.
- ✓ Check CPU, Memory and Storage needs.
- ✓ Start with smaller type and scale when needed.
- ✓ Use monitoring (CloudWatch) to analyze usage.
- ✓ Choose cost-effective option for your workload.



6. Purchasing Options

Option	Description	Best For
On-Demand	Pay by hour/second, no long term commitment.	Short term / Unpredictable workloads
Reserved	1-3 years commitment, low cost.	Steady state workloads
Spot	Unused capacity, very low cost.	Flexible, fault tolerant jobs
Savings Plans	Commit to usage (\$/hr), get lower cost.	Long term workloads

7. Key Takeaways

- ★ Instance type affects performance and cost.
- ★ Choose type based on workload, not guess.
- ★ Use monitoring to right-size your instances.
- ★ Combine with ASG for automatic scaling.

8. Quick Summary



Right Instance Type = Better Performance
Lower Cost
Scalable Application

Choose Smart . Run Efficient . Build Scalable on AWS





EC2 PRICING

14

EC2 pricing is based on how you use, when you use and what type of instance you choose.



1. Pricing Depends On

- ☆ Instance type (CPU, Memory, Storage, etc.)
- ☆ Region (each region has different price)
- ☆ Operating System (Linux is cheaper than Windows)
- ☆ Tenure / Commitment (On-Demand, Reserved, Spot, Savings Plans)
- ☆ Usage (per second billing - pay only for what you use)



You only pay for running instances. **Stopped** instances cost nothing for compute (but EBS cost may apply).

2. EC2 Pricing Models

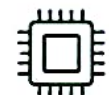
	On-Demand	Pay by the second or hour. No upfront cost. Flexible.
	Reserved Instances (RI)	Commit for 1 or 3 years. Big discount (up to 72%).
	Savings Plans	Commit to \$/hour for 1 or 3 years. Flexible across instance types.
	Spot Instances	Use spare AWS capacity. Huge discount (up to 90%). Can be interrupted.
	Dedicated Hosts	Physical server dedicated to you. Use your own licenses.

3. Example: On-Demand Pricing (ap-south-1)

Instance Type	vCPU	RAM	On-Demand Price (USD per Hour)
t3.micro	2	1 GB	\$0.0136
t3.small	2	2 GB	\$0.0272
t3.medium	2	4 GB	\$0.0544
m5.large	2	8 GB	\$0.0960
c5.large	2	4 GB	\$0.0850
r5.large	2	16 GB	\$0.1330

★ Prices change over time. Always check AWS Pricing Calculator for latest price.

4. Example: t3.micro On-Demand Cost



t3.micro

- Price: \$0.0136 per hour
- Used for: 730 hours (24x7 for 1 month)
- Cost = 0.0136×730
= \$9.93 per month (approx)

5. Pricing Model Comparison (Example: t3.micro)

Model	Commitment	Discount	Best For
On-Demand	No commitment	0%	Short term, testing
Reserved (1 Year)	1 Year	~35%	Steady workloads
Reserved (3 Year)	3 Year	~60%	Long term, stable
Savings Plan (1 Year)	1 Year (\$/hr)	~20-40%	Flexible workloads
Savings Plan (3 Year)	3 Year (\$/hr)	~30-50%	Long term, flexible
Spot Instances	No commitment	Up to 90%	Fault tolerant jobs

6. Spot Instance Pricing



Spot Instance

- Uses spare AWS capacity.
- Price changes based on supply & demand.
- You set a **Max Price** you are willing to pay.
- If market price > your max price, instance can be interrupted.
- Great for batch jobs, CI/CD, big data, machine learning, etc.

7. Other Costs to Know

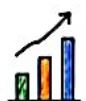
Component	What You Pay For
EBS Volume	Storage (per GB per month) Plus IOPS (for gp3/io1/io2)
Data Transfer	Data out to internet or between regions
Elastic IP	If not associated with a running instance
Snapshots	Stored in S3 (per GB per month)
Load Balancer	Per hour + data processed



Total cost = EC2 + EBS + Data Transfer + Other services
Always monitor usage with AWS Cost Explorer & Budgets.

8. Key Takeaways

- ✓ Choose the right pricing model for your workload.
- ✓ Reserve or commit for long term to save more.
- ✓ Use Spot for fault tolerant & cost-sensitive jobs.
- ✓ Always monitor usage and set budgets to control cost.



9. Quick Summary

Choose Right Instance Type → Choose Right Pricing Model → Monitor & Optimize Costs → Save More



Right Pricing Choice = Lower Cost + Better Performance + Happy Users





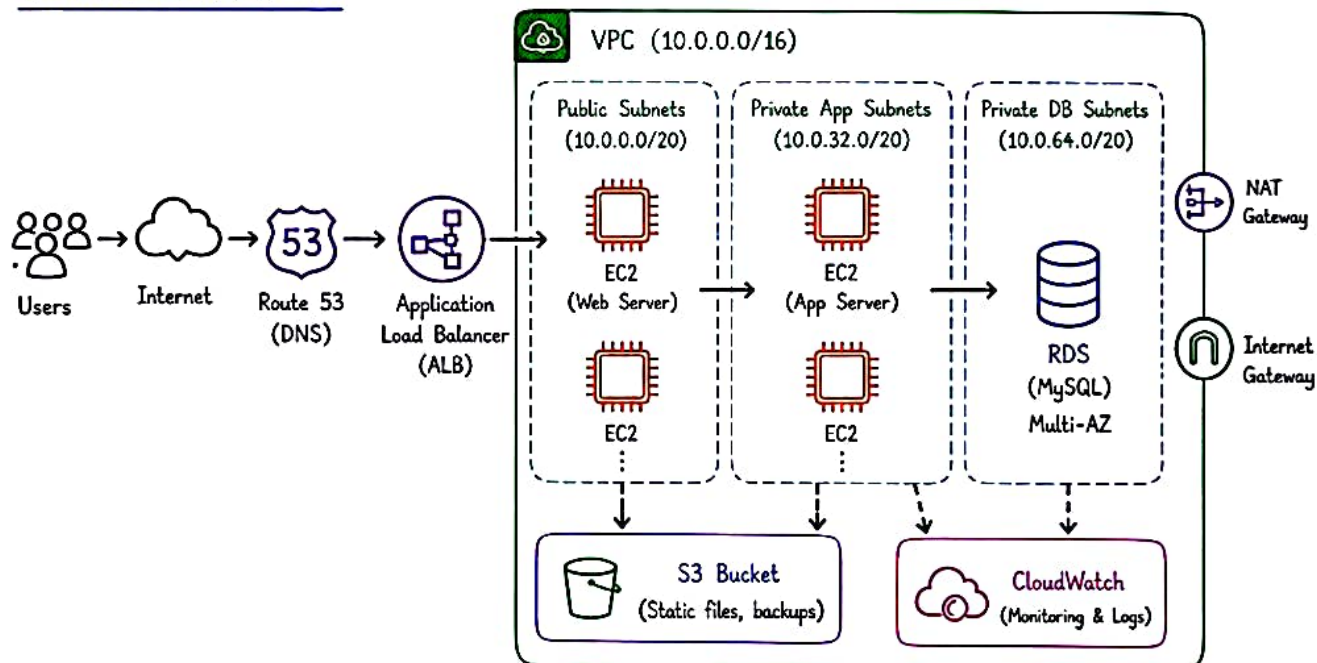
EC2 REAL-WORLD EC2 ARCHITECTURE + EXAMPLE

15



EC2 is usually used as part of a bigger architecture.
Here is a real-world example for a 3-tier Web Application.

1. Architecture Overview



2. Key Components

	Route 53	Manages DNS and routes traffic to the application.
	ALB	Distributes incoming traffic across multiple EC2 web servers.
	EC2 Web	Handles HTTP/HTTPS requests, serves static content, forwards dynamic requests to app server.
	EC2 App	Contains business logic, connects to database, processes requests.
	RDS MySQL	Stores application data. Multi-AZ for high availability.
	S3	Stores static files (images, videos, documents) and backups.
	CloudWatch	Collects metrics, logs and sets alarms.
	NAT Gateway	Allows private instances to access internet for updates, packages etc.

3. How Traffic Flows

- 1 User enters domain name → Route 53 resolves to ALB.
- 2 ALB receives the request and sends it to EC2 Web servers in public subnets.
- 3 Web server serves static content from S3 (if needed).
- 4 For dynamic requests, Web server forwards request to EC2 App servers in private subnets.
- 5 App server processes the request and reads/writes data from RDS MySQL.
- 6 Response goes back → App Server → Web Server → ALB → User.
- 7 CloudWatch monitors all resources and sends alerts if any issue occurs.

4. Benefits of This Architecture

- ✓ High Availability → Multi-AZ RDS, ALB across AZs.
- ✓ Scalability → Add more EC2 in ASG as traffic grows.
- ✓ Security → Private subnets for App & DB.
- ✓ Performance → ALB, S3, and caching improve speed.
- ✓ Monitoring → CloudWatch provides insights and alerts.

5. Example Use Case



You are building an E-Commerce website.

- Users browse products (Web tier)
- Add to cart, place order (App tier)
- Order details stored in RDS (DB tier)
- Product images stored in S3
- All servers monitored in CloudWatch



EC2 is powerful on its own, but becomes more powerful when combined with other AWS services!





EC2 TROUBLESHOOTING

+ IMPORTANT INTERVIEW CONCEPTS

16



Problems are normal.

The key is to troubleshoot in the right way and stay calm!

1. Common EC2 Issues & How to Fix

Issue	Possible Cause	What to Check / Fix
Instance not reachable 	- Security Group blocking - NACL blocking - Wrong IP / DNS - Instance stopped	- Check SG inbound rules - Check NACL rules - Verify Public IP / DNS - Check instance state
SSH Connection Timed Out 	- SG not allowing port 22 - Incorrect key pair - NACL blocking - Instance rebooting	- Allow port 22 in SG - Use correct key (.pem) - Check NACL rules - Wait and try again
Instance Reachable but App not working 	- App not running - Wrong port - Firewall on EC2 - Service crashed	- Check app/service status - Verify app port - Check OS firewall (ufw/iptables) - Check logs (app logs)
High CPU / High Memory 	- High traffic - App bug / leak - Background process	- Check with CloudWatch metrics - Identify heavy process (top) - Optimize or scale
Disk Full 	- Logs too big - Temp files - No cleanup	- Check disk (df -h) - Clean up logs/temp - Increase disk (EBS)
Instance Stuck in Pending 	- Capacity issue - Wrong AMI / config - AZ outage	- Check EC2 events - Try another AZ / type - Wait or open support case

2. Troubleshooting Checklist



1. Is the instance running?

Check status check (System & Instance).



2. Can you ping the Public IP?

If yes → Network is OK.

If no → Check SG / NACL / Route.



3. Can you SSH to the instance?

If no → Check SG (port 22) & Key.



4. Is the App / Service running?

Check service status and port.



5. Check Logs

/var/log/, app logs, CloudWatch logs.



6. Check Metrics

Use CloudWatch for CPU, Memory, Disk, Network.



7. Fix the issue and Test again!

3. Useful AWS Tools for Troubleshooting

	CloudWatch	Monitor metrics, logs and set alarms.
	EC2 Console	View instance status, events and configurations.
	VPC Flow Logs	Helps in network connectivity troubleshooting.
	AWS Systems Manager (SSM)	Connect to instances securely (no need for SSH).
	CloudTrail	Track API calls and actions for debugging.

4. Important Interview Concepts

- ★ EC2 is IaaS (Infrastructure as a Service) in AWS.
- ★ AMI is like a template. It contains OS, apps and settings.
- ★ Instance = Running server created from AMI.
- ★ Key Pair is used for SSH access to Linux instances.
- ★ Security Group works at Instance level (stateful).
- ★ NACL works at Subnet level (stateless).
- ★ EBS is block storage. Data remains even if instance stops.
- ★ Instance Store is temporary storage. Data deleted on stop/terminate.
- ★ You pay only for what you use (On-Demand Billing).
- ★ Use ASG for auto scaling and High Availability.
- ★ Always keep your AMI updated and secure.

5. Common Interview Questions

- What is EC2?
- What is the difference between Instance Store and EBS?
- What is the difference between Security Group and NACL?
- What is an AMI?
- What is Auto Scaling Group?
- How does EC2 pricing work?
- How do you connect to an EC2 Linux instance?
- What is the default port for SSH?
- What is the purpose of Elastic IP?
- What happens if you stop vs terminate an instance?



Understand. Practice. Troubleshoot. That's how you become a Great DevOps Engineer!

